Check for updates

ORIGINAL ARTICLE

EMERGING RISK FACTORS ASSOCIATED WITH PREVALENCE OF HEPATITIS C VIRUS INFECTION AMONG NIGERIANS: FINDINGS FROM BLOOD DONORS IN AN ACADEMIC HOSPITAL, ENUGU SOUTHEASTERN NIGERIA

Edwin Nkemjika Okafor [1], Innocent N. Okonkwo[1], Martin C.Ugonabo[2], Ekene E. Chukwukelu[2], Obiageli U. Odurukwe [3], N. S. Osiri[4]

[1]Department of Medical Laboratory Sciences, University of Nigeria, Enugu Campus.

[2]Department of Chemical Pathology, Faculty of Medical Sciences, University of Nigeria Enugu Campus.

[3]Department of Chemical Pathology, University of Nigeria Teaching Hospital Enugu.

[4]Department of Haematology and Blood Transfusion Services, University of Nigeria Teaching Hospital, Enugu

Corresponding Author:

Dr. Edwin Nkemjika Okafor, Department of Medical Laboratory Sciences, University of Nigeria, Enugu Campus

Email: nkemjika.okafor@unn.edu.ng

Running Title: Risk Factors of Hepatitis C Infection

Accepted Article

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/IJCP.13460

This article is protected by copyright. All rights reserved

Acknowledgement

This is part of research program from department of Chemical Pathology, Haematology and Blood Transfusion Services, University of Nigeria Teaching Hospital Enugu, Nigeria to create awareness for screening of risk factors as a result of increasing prevalence of Hepatitis C virus infection in Nigeria.

Author's Contributions

This work was carried out in collaboration between all authors. Authors: Edwin N Okafor, Ugonabo Martin C and Ekene Chukwukelu E, deigned the study, wrote the protocol and wrote the first draft of the manuscript. Authors Odurukwe O. U., Innocent Okonkwo N., managed the literature searches. Osiri N. S. Edwin N Okafor, managed the experimental process and did statistical analysis. All authors read the approved manuscript for submission to International Journal of Clinical Practice.

Conflict of interest Nil

Financial Support and Sponsorship Nil

Accepted Article

This article is protected by copyright. All rights reserved

Accepted Article

DR EDWIN NKEMJIKA OKAFOR (Orcid ID : 0000-0001-6795-603X)

Article type : Original Paper

ABSTRACT

Aim: In Nigeria, reports on the prevalence of Hepatitis C risk factors have not been clearly elucidated, we investigate the risk factors that influence the prevalence and how the difference in awareness level of risk factors screening across age groups contributes to the disease.

Methods: A total of 8790 aged 18-55 years old, who attended the Department of Haematology and Blood Transfusion Services, University of Nigeria Teaching Hospital, Enugu were screened. Among them 135 were diagnosed with Hepatitis C infection from January 1, 2017 – January 1, 2019. Participants completed a questionnaire related to demographics, risk factors, history of previous medical screening. Age groups, income and education were indicators. Multivariate analyses were used to identify correlates.

Results: A total of 135 blood donors (95 males and 40 females) with a mean age of 35.9±10.5 and 36.4±10.7 years respectively were positive for Hepatitis C infection, giving overall prevalence (1.5%), injecting drug use (39%), unprotected multiple sex (22.2%), sacrification (14.8%), blood transfusion (11.1%). Multivariate analysis revealed that injecting drug use (*P<0.0001)*, unprotected multiple sex (P=0.004), sacrification (P=0.036) were independent risk factors that influence the prevalence among younger age groups. Blood transfusion was not significant *(P=0.053)* but the mostly accessed. Low income and educational background were significantly more prevalent among the studied group.

Conclusion: Injecting drug use, multiple sex, and sacrification are independent predictors of prevalence of Hepatitis C infection. The early onset of the risk factors among the younger adult clearly underlines the need for early screening and interventions.

This article is protected by copyright. All rights reserved

Accepted Article

- Risk factors for HCV infection have been documented. A recent report published in Enugu concluded that blood transfusion might not be the only main route of transmission of HCV infection as a result of increasing prevalence in Nigeria, that there are other risk factors yet to be fully defined. We investigate the likelihood of injecting drug use, sacrification and multiple sex as HCV risk behavior which have not been clearly defined in Nigerian population compared to other developed countries.

What's already known about this topic?

Risk factors for HCV infection have been documented. A recent report published in Enugu concluded that blood transfusion might not be the only main route of transmission of HCV infection as a result of increasing prevalence in Nigeria, that there are other risk factors yet to be full defined.

What does this article add?

We investigate the likelihood of injecting drug use, sacrification and multiple sex as HCV risk behavior which have not been clearly defined in Nigerian population compared to other developed countries.

Key Words: Prevalence, Risk Factors, Hepatitis C Virus Antibodies, Nigeria

This article is protected by copyright. All rights reserved

Accepted Article

Introduction

The prevalence of hepatitis C infection has increased steadily in the past years, impacting on hundreds of thousands of Nigerians[1.] Hepatitis C, caused by hepatitis C virus (HCV) is a global health problem. According to the World Health Organization (WHO), there are at least 3-4 million people infected annually with HCV, 80% of whom progress to chronic disease.[2] In addition to end stage liver disease (ESLD), hepatocellular carcinoma (HCC) is the important sequela of chronic HCV infections.[3] Provisional data suggested the occurrence of 782,000 fatalities due to primary liver cancer in 2018. Among these deaths, an estimated 664,700 deaths involved HCC[4].

Available data showed that the global prevalence of HCV is 2.5%[5]. In Nigeria, studies conducted in various population subgroups like sickle cell disease (SCD), and blood donors (BDs) showed different prevalence rates ranging from 0.4% to 14%[6,7]. Some risk factors associated with HCV infection are blood transfusion or blood products, tattooing with re-usage of instruments, body piercing, injection/illicit drug users, multiple homosexual and heterosexual partners as well as perinatal exposure to infected body fluid[2]. HCV and its associated complications have a significant impact on patients' quality of life. The magnitudes of the effects are diverse. A 2012 retrospective survey involving 401 Nigerian HIV positive pregnant women at the University of Nigeria Teaching Hospital (UNTH), Enugu showed that co-infection of HCV and Hepatitis B surface Antigen (HbsAg) had a sizeable prevalence of 6.5%[8]. HCV co-infection has been found to be more common in HIV positive individuals and is associated with the risk of antiretroviral drug induced hepatotoxicity[9], increased mortality and renal morbidity[10]. HCV is transmitted efficiently by large exposure to blood such as transfusion of blood. Blood transfusion, a common medical intervention to correct anemia in high risk patient is effective and endorsed by prominent guideline[11].However, inclusive testing of antibodies to HCV is not a routine practice even among high risk population like blood donors, thus making the pathogen still a neglected one. Therefore, there is need for renewed interest in the risk factors as report has shown uprising in the prevalence of HCV infections[1]. A case control study among British blood donors found that having been healthcare worker was a risk factor for having HCV infection[12].

University of Nigeria Teaching Hospital (UNTH) Enugu has an ongoing screening program for infectious disease control. The program protocol involves opportunistic screening for HCV infection among individuals ≥ 18 years of age attending the hospital for blood donation. The program is on its early stages of implementation and preliminary assessment suggested little data regarding the prevalence of risk factors. Majority of previous studies

This article is protected by copyright. All rights reserved

focused on blood transfusion as the major route of transmission of this virus.[13,14].Report documented that blood transfusion does not appear to be the only major route influencing the prevalence of HCV transmission in Enugu, as there are others yet to be fully defined[14]. This is an important gap in research. Although report of outbreak of HIV/HCV infection was linked to injection/syringe sharing in USA,[15] but this scenario is yet to be defined. In this study, our goal was to identify the risk factors associated with the prevalence of HCV infection. We examine the following research questions: (1) Risk factors associated with increased likelihood of HCV prevalence (2) Does the effect of HCV prevalence vary by the respondent age group income, education on level of screening awareness.

Accepted Article

This article is protected by copyright. All rights reserved

Accepted Article

MATERIAL AND METHODS

Study site and design

This was a prospective study with multiple prevention strategies. The period was January 1st 2017 through January 1st 2019. University of Nigeria Teaching Hospital (UNTH) Enugu is a referral academic hospital, with network of 4 outposts that serves mainly southeastern states of Nigeria. It is located in Enugu State of Nigeria. It is a 700 bed capacity hospital. It runs well established blood banking services which care for the patients. This site was selected because of the highest number of individuals attending the hospital for blood donation. We also had the highest number of blood donors involved in injecting drug. Strategic collaboration includes multi-professional teams from medicine, chemical pathology and blood transfusion services in conjunction with local health department. Before commencement with the study, we engaged in discussion with blood donors in disclosing personal information. From these conversations, we discovered that these blood donors were most receptive to our study when data were anonymously. Furthermore, we discovered that people may be reluctant to disclose any unusual behavioral risk e.g. drug of abuse because of stigma. As a result we set out inclusion criteria. To be at least 18 years old with no clinical evidence of any other chronic disease and verifiable contact address. Approval was obtained from UNTH ethical committee before commencement of this study. Informed written consent was obtained from each participant.

Participant for enrollment

A total of 8,790 blood donors were screened within the period of study and of which, 136 individuals aged 18-55 years were identified as seropositive to HCV infection. The diagnosis was based on new (second) generation HCV one step test strip (serum/plasma from acumen diagnostic INC. USA) [16]. The sensitivity of the test strip is > 99% while its specificity is 98.6% [16]. Active RNA infection of all seropositive samples was confirmed with HCV RNA testing[17]. Detection of viremia was recorded as target detected or target not detected. All those positive in RNA testing was eligible for inclusion in the study. 136 individuals met the case definition and were invited to questionnaire completion of whom 135 (99.3%) attended .One had thyroid dysfunction and was excluded for expert management in the hospital.

Demographics and other questionnaire variables

This article is protected by copyright. All rights reserved

All 135 participants completed a questionnaire under the guidance of medical staff and were given opportunity to ask questions. The information included a diversity of measures including demographics data as well as items relevant to HCV infection (injecting drug use HCV and HIV risk behavior (eg. syringe sharing (SS)), blood transfusion, scarification/tattooing, unprotected multiple sex) and history of medical screening. Demographics including, age in years, highest level of education completed and employment status. Average income earnings were self reported and recorded in the questionnaire during a face to face interview. Average income earnings were reported in Nigeria currency (Naira) but were converted to International currency via Google on 30th July, 2019. (₦1= US 0.0061) and categorized into groups, low income group (<$108/month), middle income ($109- $309 / month), upper middle income ($310-$644/month) and high income (> $644/month).

History of Previous Medical Checkup/Screening and Awareness

Participant's accessibility to HCV risk screening/awareness was reported during the face-to-face interview. Participants were asked whether or not they have ever had the knowledge of HCV risk factors or screened by a doctor or other health care workers Responses were recorded in the questionnaires.

Statistical Analyses

We calculated the prevalence data in various age groups within 95th percentile. Frequency table for the demographic information were determined. Prevalence of HCV risk factors was determined. All data are presented as mean ± SD for continuous variables or as numbers and percentages for categorical variables. The independent t test was used to identify difference between continuous variables. Multiple logistics was applied to access the contribution of the risk factors to HCV infection prevalence. The SPSS ver. 16 (SPSS Inc. Chicago) was used for statistical analysis $p< 0.05$ were considered to indicate significant.

This article is protected by copyright. All rights reserved

Accepted Article

RESULTS

Baseline Characteristics of the Participants and Average Mean ± SD of Demographics

The baseline characteristics of all HCV positive participants are listed in table 1.A total of 135 participants aged 18-55 years were enrolled according to study criteria, of whom 95(70.4%) were male. The mean age was 35.9±10.5 for male and 36.4±10.5 for female, but there was no significant (P>0.05) difference between their mean ages. The average daily income of male and female was $19.50±4.85 and $7.39±1.44 respectively, but significant (P<0.05) difference exist between their average daily income. The log viral load of male 4.59 ± 0.16 was not significant (P>0.05) compared to female 4.59 ± 0.1.

In this study population, overall prevalence was (1.5%). The most reported age group being the younger age group less than 40 years 18 – 29 years (53.7%), 30 – 39 years (29.4%). Additionally, 22.2% reported no formal education, 25.9% less than primary education, 33.2% reported Primary school, 14.8% reported Secondary school while 3.7% University education. Their current job varied, with unskilled workers (44.4%), skilled workers (7.4%), and unemployed (48.1%). Average daily income also varied, low income (59.3%), low middle income (14.8%), upper middle income (3.0%), high income (0.7%), and unknown (22.2%). According to the result, the most occurring risk factor was injecting drug use (39%), .It was followed by unprotected multiple sexual partners (22.2%), sacrification/tattooing (14.8%), blood transfusion (11.1%), accidental needle injury (2.2%), peri-natal (0.7%). (Table I)

The result showed that lower age groups 18 – 29 and 30 – 39 years had significant (P<0.0001) prevalence of injecting drug use 54.8% (95% Cl: 35–65), 37.5% (95% Cl: 30–40) respectively. But screening awareness was not significant (*P=0.459*) among that age group. Also prevalence of multiple sex partners exist in lower age group with highest prevalence seen to occur at 40 -49 years old participants with 33.5%(95% Cl: 25-39). However awareness was significant (*P=0.002*) increased with increasing age peaking at > 50 years A significant prevalence of sacrification/tattooing 32.5%(95% Cl: 26.1 39.5) was seen at age group 30-39 years and this is significant (*P<0.0001*). Across all age groups mostly accessed HCV risk screening was blood transfusion (X^2, P <0.0001) with a peak at 18-29 years old participant 58.9%(95% Cl: 40 -70) while the least accessed was injecting drug use. (Table II). There was a significant difference between individual with low income, low middle income in the percentage of risk factors than upper income and higher income. There was a

This article is protected by copyright. All rights reserved

significant difference between individual with university, secondary school than no formal education, and less than primary education. (Table III).

Accepted Article

This article is protected by copyright. All rights reserved

Accepted Article

Discussion

The prevalence of HCV infection has increased steadily in Nigeria, it was estimated by 2010, that the prevalence had reached 6.6% in Enugu Southeastern Nigeria,[1] creating opportunity for HCV-associated complications. The most common and bothersome complication is hepatocelluar carcinoma, which caused 85% of all liver cancer in 2018[4]. This may cause irreversible changes to the liver function leading to more fatalities. This however may not signify improved hepatic care and control of risk factors for liver cancers. Therefore, it is important to explore the risk factors associated to influence the prevalence of Hepatitis C infection, in order to stop the spread, facilitate early detection for a better prognosis.

The results from this study indicate that blood donors attending academic hospital Enugu bear substantial risk factors of HCV infection. Their major risk factors are injecting drug/syringe sharing, multiple sex partners and sacrification/tattooing. These variables were present in more significant proportions. This calls for a tailored program to reduce the risk. A report showed that risk factors interventions, (harm reduction counseling and education) may lower the risk factor of injecting drug[1]. However, strategies to improve the prevention and early detection of HCV in low and middle income countries like Nigeria are not only hampered by economic consideration, but also limited data emerging from these countries. These countries have witnessed rapid deterioration of their chronic disease risk and mortality profiles[18]. In the multivariate analysis, three risk factors were found to be independent risk factor for the prevalence of HCV infection. Taken together, we can infer that the risks of being infected increase with the likelihood of these factors, injecting drug use/syringe sharing multiple sex partners and sacrification/tattooing. Blood transfusion was the least occurring risk factor but most accessed. A previous report recorded blood transfusion in Ogbomosho Southwestern Nigeria as the common risk factor[19]. This indicates that high risk of exposure, socio-demographic and lifestyle changes may have contributed to the variation in the frequency of these risk factors. This can be explained why this present work and that of Ejiofor *et al* which was done in Enugu Southeastern Nigeria recorded sacrification as significant risk factor[20]. .However, Atshaha *et al* in Ethiopia recorded history of hospitalization, tooth extraction and blood transfusion[21]. The common cluster of risk factors (unprotected multiple sex, sacrification/tattooing, blood transfusion) recorded in this study is a common feature of most studies in Nigeria[19] and some part of Africa[22]. This could be explained by characteristics of African society where sacrifications are embedded in their custom/lifestyle for the service or worship of a deity.

This article is protected by copyright. All rights reserved

Accepted Article

Injecting drug use is an emerging risk factor which was considered to be rare among Nigerians, but has been recorded in London 34 years ago[12] .The reverse is the case in present time. This research fills an important gap in the risk assessment of blood donors in Nigeria as scanty population level data exist, particularly among blood donors identified as vulnerable to HCV related infection As age of the subject differ, it indicate that the prevalence was explained by age factor since it was stratified by age, the study demonstrates that injecting drug use is likely the mode of transmission of over 54.8% of HCV infection in younger age group of 18- 29 years. In a research study documented in Scott county Indiana in United states of America, (USA), it was found that 92% of new cases of HCV/HIV was linked to injecting drug/syringe sharing[23].As the injecting drug use is not included in the risk assessment procedure for potential blood donors in our hospitals, it is likely that injecting drug use is responsible for influencing the prevalence of HCV infection. A higher percentage of injecting drug was reported among HCV positive donors in London many years ago[12].This is the reason why injecting drug should be considered as a major factor in the transmission of HCV infection in Nigerian population. Little awareness was given to HCV screening, so it is unlikely that the donors gave blood to obtain Hepatitis C test. More cases were on the regular donor panel. Therefore, this risk factor can be used to reflex the prevalence of HCV infection. It is imperative that the healthcare facilities implement strategies that are scaled to meet population level needs, so that the gains from the control of protracted HIV/AIDS in Nigeria may not be reversed by another outbreak of HCV infection. These data also have significant implications for HIV prevention initiatives, since both HIV and HCV share the same route of transmission. Additionally, about 70% of the donors reside in rural communities, creating opportunity for an increase in community level vulnerability to outbreak of HCV as research has shown that the intravenous drug users who report injecting drug are most likely to inject with greater frequency and re-usage of syringe[24.] Unfortunately, there are no up-to-date information regarding the size and characteristics of these blood donors that are involved in this abuse. This can creates significant challenges for the centre for disease control and prevention (CDC).Our analysis can help to identify those rural communities with highest risk factor in the spread of HCV infection where intervention can be prioritized. Targeted screening to identify individuals with high risk characteristics should be undertaken. A variety of population estimation methodologies can be used to estimate the size of vulnerable population.[25]It was reported that high prevalence in injecting drug use was notable for the absence of harm-reduction strategies associated with HCV infection such as addiction treatment, rehabilitation, medication assisted therapy and syringe access programs in U.S.A[15].

This article is protected by copyright. All rights reserved

This demonstrates the need for the harm reduction strategies to be situated in our population where majority of the donors involved in drug reside.

Multiple sex partners cannot be protective against HCV infection, this finding is evidence against significant sexual transmission of Hepatitis C. This agrees with a study[26]. While being contrary to the other[2.] Our study reported that the percentage prevalence of multiple sex was at its peak among donors aged 40- 49 years. This may be explained with the fact that most of the HCV positive donors are male and at this particular age group, they are sexually active and in a better position to keep as many sexual partners as possible. The social behaviour in this part of the world allows polygamy because men are better financially established in this particular period in their life. In a study, it was found that increased risk of HCV infection in individuals with other sexually transmitted infection including HIV was largely driven by co-existence by HCV multiple risk factors[8], as was observed in our study. This becomes alarming for a setting where algorithms for early screening and detection of disease risk factors are under-assessed. It is argued that the effect of anti-viral therapy is to achieve a sustained virologic reponse,[27] and this can delay complication of the disease, but this can be the case in societies with effective health awareness and health seeking behavior that will enhance the opportunity for early detection and intervention. Many sexual act are often use as an indicator of risky sexual behavior and many interventions are designed to deliver abstinence education[28]. This method will not be sufficient strategies. Given that HCV risk represent a serious adverse effect among younger adults, strategies such as providing healthcare and sex education to youth before they are sexually active should be encouraged.

Association with tattooing and direct evidence to support tattooing has been documented[22]. In our report, the highest prevalence was seen at 30-39, this age groups reflects maximum initiation for the service of a deity. Our findings collaborated with the report documented in Enugu Southeastern Nigeria[13]. Report of people tattooing themselves or by the traditional herbalist in Nigerian culture with unsterilized surgical material or needle support our finding.

Interestingly, blood transfusion showed the least risk factor. It was not found to be an independent risk factor. Although blood transfusion was previously reported as the most common risk factor in the transmission of HCV[13,14]. It was not suitable in evaluating the prevalence of HCV infection but most accessed. This could be due to the evidence that the rate of exposure for blood transfusion has decreased overtime temporary, associated with increase awareness of the risk involved in blood transfusion and compliance with the practice

This article is protected by copyright. All rights reserved

Accepted Article

Accepted Article

of standard precautions. Secondly, extensive use of recombinant erthyropoietin for anemia correction has markedly reduced the need for transfusion. However, a detailed risk assessment for HCV infection should include blood transfusion. Our study demonstrates insignificant risk of accidental injury .To minimize the risk, all including the health workers should adhere to standard precaution, including appropriate use of hand washing, protective barriers and care in the use of disposal of needle and other sharp instrument. Health workers should have a system that includes written protocols for prompt reporting, evaluation, counseling and follow up of occupational exposure that may placed the worker at risk of HCV infection. This is because, despite improve methods of preventing exposure, occupational exposure will continue to occur. It is of concern that people who have engaged in high-risk patterns of behavior still donate blood. An explanation for this failure of donors with a history of high-risk behavior to exclude themselves was that the self-exclusion criteria were in the information aimed at preventing HIV transmission.

The socioeconomic factors across the study population show that majority of blood donors are more disadvantaged in terms of income earning. We noted that HCV infections were found more in lower socioeconomic class than other social classes. Our finding is corroborated with a study carried out in Southeastern Nigeria[20]. The greater percentage of donors is living in poverty, which the WHO defined in absolute terms as low income less than $2/day. The minimum wage in Nigeria is ₦18,000/month which is approximately $108/month and categorized as low income level .This may be the reason why the study population rely on the error-prone antibody method of testing which is affordable. The cause of RNA testing of HCV in Nigeria is equal to $200 which is above the minimum wage of a Nigerian worker.

Again this study showed that education is a protective factor against HCV infection. The trend shows that individuals with less than primary school education had higher prevalence of HCV risk factors while those with university education have low prevalence. This may be due to increased awareness and knowledge of the dangers of contracting the infection and also behavioral change in lifestyle. It is important to lobby companies, educational institutions, government and non-governmental organization to engage in health education program to enhance awareness and health seeking awareness across the population. The study identifies disparity in the accessibility to HCV risk factor screening awareness. This highlights the consequences of health disparity whereby more affluent in society have better access to health care[29].We also noted that the participant cite various reasons for not seeking

This article is protected by copyright. All rights reserved

health check [30],including cost which does affect people habit where they do not seek health check when they are apparently healthy. Therefore, attitudinal changes which impart positively on health outcome remain one of the public health messages that cannot be avoided

Limitation

It is a single centre research, the result might not represent all HCV individuals therefore, cannot be generalized. The main source of bias is response bias. Cases were interviewed after they were tested positive for HCV infection so it is possible that the participants responded more truthfully. The potential response bias should probably be greatest for the history of injecting drug and for sexual histories. Recall bias should be low when the donors are given time to answer the questions and it is also unlikely that people forget that they have injected drugs, tattooed or sexual partner.

Conclusion

The prevalence of HCV infection is associated with injecting drug use, multiple sex, and sacrification, income and education. Injecting drug use explains higher percentage of hepatitis C virus infection among the blood donors. Designing national effective intervention programs is necessary.

Accepted Article

This article is protected by copyright. All rights reserved

Accepted Article

REFERENCES

1. Ejiofor S.O, Emechebe G.O. Igwe W.C *et al*. Hepatitis C virus infection in Nigeria. Niger Med J 2010; 51(4):173-176.
2. WHO. Hepatitis C Fact sheet N°164.2000 [updated 2000 July 2012; cited]. Available from: http://www.who.int/mediacentre/factsheets/fs164/en/.
3. Bernstern D, Diagnosis and the management of Hepatitis C available at: http://www.medscape.com/medscape/clinical/mgnt/cm.vol/public/indexcm.vol.html.December 2008.
4. Bray F, Ferlay J, Soerjomataram I., *et al*. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2018 Nov; 68(6):394-424. doi: 10.3322/caac.21492. Epub 2018 Sep 12.
5. Petruzziello A, Marigliano S, Loquercio G, *et al*. Global epidemiology of hepatitis C virus Infection: An up-date of the distribution and circulation of hepatitis C virus genotypes. World J Gastroenterol. 2016;22(34):7824-7840. doi 10.3748/wjg.v22.i34.7824.
6. Nwannadi I. A., Alao, O. O., Bazuaye, G. N., *et al*. Seroprevalence of Hepatitis C Virus Antibody in Sickle Cell Anemia Patients in Benin City, Nigeria. Gomal J of Med Sci. 2012;10:1.
7. Udeze A. O, Okonkwo I. O, Donbrye E, *et al*. Seropreevalence of Hepatitis C virus Antibodies Among Blood Donors in Ibadan, Southwestern, Nigeria. World Appl. Sci. 2009;7:1023-1028.
8. Okeke TC, Obi SN, Okezie OA, *et al*. Infection with Hepatitis B and C Viruses and HIV Positive Pregnant Women in Enugu South East, Nigeria. Nigeria Journal of Medicine. 2012;21(1).
9. Greub G, Clinical Prevalence, survival and immune recovery during antiretroviral therapy in patient with HIV-1 and Hepatitis C Co-infection , SWISS HIV Cohort Study Lancet. 200;356:1800-1805.
10. Thomas DL. Hepatitis C and Human Immunodeficiency Virus Infection. Hepatology. 2002 Nov, 36(5Suppli):5201-9.

This article is protected by copyright. All rights reserved

Accepted Article

11. Jayanthi A. Zero adverse events. How Dignity Health achieved new standard. Becker's Hospital Review: Infection Control and Clinical Quality. Available at: http://www.beckershospitalreview.com/quality/zero-adverse-how-dignity-health-achieved-a-new-standard.html. accessed April 19, 2016.

12. Neal K R, Jones DA, Killey D, *et al*. Risk Factors for hepatitis c virus infection : a case control study of blood donors in the Trent Region (UK). Epidemiol infect 1994;112:595-601

13. Ejiofor OS, Ibe BC, Emodi IJ, *et al*. The role of blood transfusion on the prevalence of hepatitis C virus antibodies in children with sickle cell anemia in Enugu, South East Nigeria. Niger J Clin. Pract. 2009 Dec;12(4):355-8.

14. Lesi OA, Kehinde MO. Hepatitis C virus infection in patients with sickle cell anemia at the Lagos University Hospital. Niger Postgrad Med J. 2003 Jun;10(2):79-83.

15. Van Handel, M M, Rose CE, Hallisey E J, *et al*. County-level Vulnerability Assessment for Rapid Dissemination of HIV or HCV Infections among Persons who Inject Drugs, United States. J Acquir. Immune Defic Syndr. 2016 Nov 1; 73(3): 323–331. doi: 10.1097/QAI.0000000000001098. PMCID: PMC5479631 **A**

16. Wilber J.C Development and Use of Laboratory Tests for Hepatitis C Infection. A Review J.Clin. Immunoassay 1993; 16:204

17. CAP/CTM HCV v2.0, "COBAS.AmpliPrep/COBAS* Taqman* HCV test

18. World Health Organization: The world health report 2002; Reducing Risks, promoting Healthy life Geneva, Switzerland. World Health Organization 2002.

19. Jemilohun AC, Bolaji OO, Sebastive OO. Prevalence of hepatitis C virus antibody among undergraduates in Ogbomosho Southwestern Nigeria. Afri. J. Infection. Dis. 2014;8(2):40-43.

20. Ejiofor S.O,Ibe B.C, Emodi IJ et al. Sociodemographic differences in the distribution of hepatitis C antibodies among children with sickle cell anemia in Enugu, Southeastern Nigeria. OJM 2009:21:50-56

21. Atsbasha A.H, Dejen T.A , Belodu R., *et al.* Voluntary Counseling testing and anti retroviral treatment clinic in Ethiopia. BMC Research Notes 2016; a(1) article 12116,2016.

This article is protected by copyright. All rights reserved

22. Esperance CL, Fabian N, John K, *et al.* Prevalence of hepatitis C virus infection and its risk factors attending Rwanda military Hospital, Rwanda Biomed Research International vol.2017, Article 105841272, http://doi.org/10.1155/2017/5841272

23. Peters PJ, Pontones P, Hoover KW et al. HIV infection linked to injection of soxymorphone in Indiana 2014-2015. N Engl J med. 2016; 375(3): 229-239

24. Centers for Disease Control and Prevention. Community Outbreak of HIV infection linked to injection use of oxymorphone-Indiana 2015. MMWR Morbmortal wkly Rep.2015,64 (16) 443-444

25. UNAIDS|WHO working group on Global HIV\AIDS and STI Surveillance Guideline in Estimating the size of population most at risk to HIV Geneva Switzerland. World Health Organization 2010:1-48

26. Eyster ME, Alter HJ, Aledort LM, *et al*. Heterosexual transmission of hepatitis C virus (HCV) and human immunodeficiency virus (HCV) . Ann Int Med 1991:115:76 4-8

27. Booth J.CL, O, Goady J, Neuberger J. Clinical Guideline on management of hepatitis C. Gut 2001;(Suppl 1): 1-21

28. Perri KK, DeJoy SB, Abstinence -only education how we get here and where we are going. J. Public Health Policy 2003;24:445-59

29. Health Disparity. http://www.niaid.nih.gov/topics/minorityhealth/pages/disparity.aspx

30. Oguoma, VM, Nwose EU, Bwititi, PT. Cardiovascular disease risk prevention: preliminary survey of baseline knowledge, attitude, and practice of a Nigerian rural community. N Am J Med Sci. 2014;6(9):466-71

Accepted Article

This article is protected by copyright. All rights reserved

Table I: Demographic Information and average mean among HCV Positive Blood Donors (n =135)

Characteristics	Values
Gender	
Male	95 (70.4%)
Age	35.9±10.5
Female	40 (29.4%)
Age	36.4±10.7
Age group years	
18-29	73 (53.7%)
30-39	40 (29.4%)
40-49	12 (8.8%)
>50	10 (7.4%)
Geographical location	
Urban	100 (74.1%)
Rural	35 (25.9%)
Education Completed	
No formal school	30 (22.2%)
Less than Primary school	35 (25.9%)
Primary school	45 (33.2%)
Secondary School	20 (14.8%)
University	5 (3.7%)
Occupation	
Unskilled	60 (44.4%)
Skilled	10 (7.4%)
Unemployed	65 (48.1%)
Average daily income (US $)	
Low income	80 (59.3%)
Low middle income	20 (14.8%)
Upper Middle income	4 (3.0%)
High income	1 (0.7%)
Unknown	30 (22.2%)
Male (average daily income)	19.5±4.85

This article is protected by copyright. All rights reserved

Accepted Article

Female(average daily income)	7.3±1.4
Behavioral risk factor	
Blood transfusion	15 (11.1%)
Injecting drug use	
Syringe sharing (SS)	*52 (39%)*
Unprotected sexual act	
Multiple sex partners	*30 (22.2%)*
Injecting drug use with sex	*10 (7.3%)*
Sacrification/tattooing	20 (14.8%)
Sacrification with injecting drug use	4 (3.0%)
Accidental needle injury	3 (2.2%)
Perinatal	1 (0.7%)

Note: Value represents mean± standard deviation or numbers (%)

Abbreviation: SS: Syringe Sharing

Accepted Article

This article is protected by copyright. All rights reserved

Prevalence of behavioral risk factors for HCV and HCV risk screening across age groups

A. HCV risks	**18-29 yrs (n=73)** (%) **95%C1**	**30-39 yrs (n=40)** (%)**95% C1**	**40-49 yrs (n=12)** (%)**95% C1**	**50 yrs (n=10)** (%)**95% C1**	**P -value**
Injecting drug use	54.8 (35-65)	37.5(30-45)	25(6-41)	40(28-45)	*<0.0001**
Multiple sex	20.5 (10-28)	17.5 (6.1-28.9)	33.3(25-39)	10(5-15)	*0.004**
Sacrification/tattooing	11.0(4-16)	32.5 (21.6-40.5)	8.3 (3.0-14)	20(10-27)	*0.036*\]*
Blood transfusion	6.8 (3-9.5)	10(5-15.6)	25(16-30)	30 (19-45)	0.053
Accidental injuries	6.8(3.8-15)	2.5(0.0-5.8)	8.3 (3.5-15)	0.00	0.398
B. History of HCV risk screening/Checkup					
Blood Transfusion	58.9 (40.6-70)	50 (44-65)	33.5(21-45)	20 (15-28)	*<0.0001***
Multiple sex partners	20.5(7-28)	25 (8-29)	25 (10-30)	50 (45-65)	0.002*
Sacrification/tattooing	13.7(5-18)	12 (6-18)	25 (10-31.5)	10 (9.5-15)	0.357
Injecting drug use (IDU)	6.8 (3-12)	12.5(6-18.5)	16.7 (3-21)	20 (15-27)	*0.459*

*P<0.05, **P<0.0001 represent statistical difference

Abbreviation: IDU: Injecting Drug Use

Accepted Article

This article is protected by copyright. All rights reserved

Table III: Percentage prevalence of HCV risk factors subgroups of average daily income and level of education among study population

HCV Risk Factors		Blood transfusion %(95% C1)	Injecting drug use %(95% C1)	Unprotected sexual act %(95% C1)	Accidental injury %(95% C1)	Sacrification %(95% C1)	P-value
Average daily income (US$)	Low income	56(37.7-63)	40(33-60.4)	80.3(64.4-100)	53.3(43-64)	40(32-45	0.0001**
	Low middle income	20(12-28)	45(34-48.6)	40.2(32.3-48.2)	49(11.9-66.7)	30(11-55)	0.0001**
	Upper middle income	6.8(2.3-12.5)	30(11.1-55.6)	19.5(13.8-28.7)	38(26.4-52.8)	21(12-55)	0.566
	High income	2.0(0.0-6.1)	1.8(0.0-4.3)	8.2(2.7-12.5)	17.5(13-25-15)	15(6-13)	0.567
Education completed	No formal education and less than primary education	55(43.0-64.1)	45(19-6.7)	60(51-78)	50(37-63)	41((41-61)	0.0001**
	Primary education	50(41.4-61.3)	40(26-55.9)	75(69-80)	47(39-59.6)	50(41-67)	0.0001**
	Secondary education	45(31`-57.4)	40(26-54)	60(51-78)	38(28-47)	25(18-35)	0.860
	University	2(0.6-6.1)	5(3-10)	5(2.5-12.5)	4(2.1-10)	3(0.0-5.0)	0.980

P<0.05 is considered Significant*

Accepted Article

This article is protected by copyright. All rights reserved